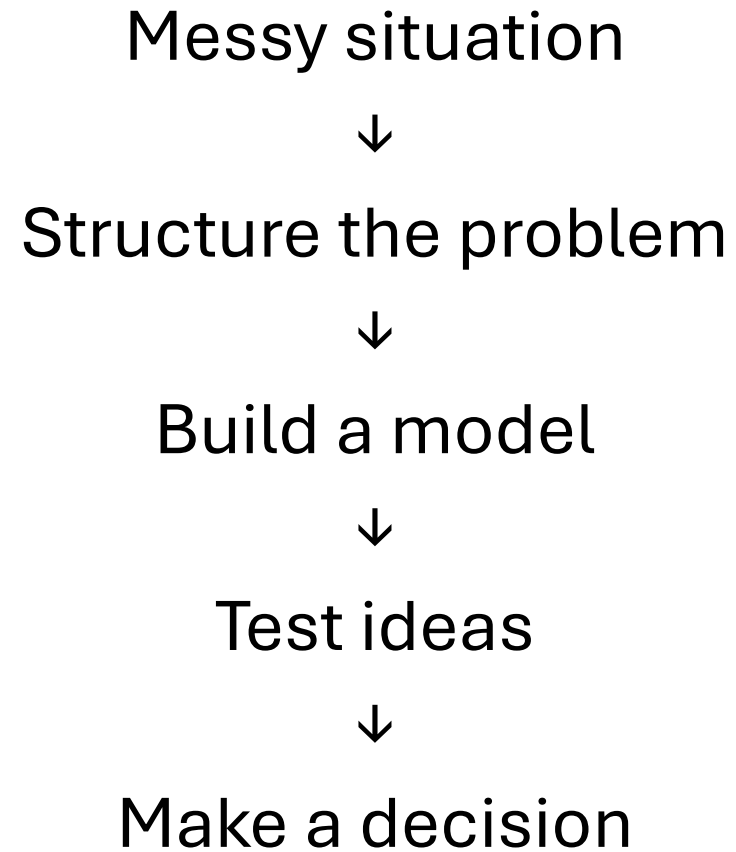


Problem Structuring and Modelling

Emergency Department

messy problem → model → decision

Session Plan



Very brief intro to “Soft OR”

Soft OR

Analytical
Thinking

*Further reading:

- Strategic Options Development and Analysis (Eden et al.)
- Soft Systems Methodology (Pidd et al.)

Presentation
skill

Toolkit of
approaches*

Working with groups

Planning

- Ownership and stakeholders
- Purpose
- Agenda

Facilitation

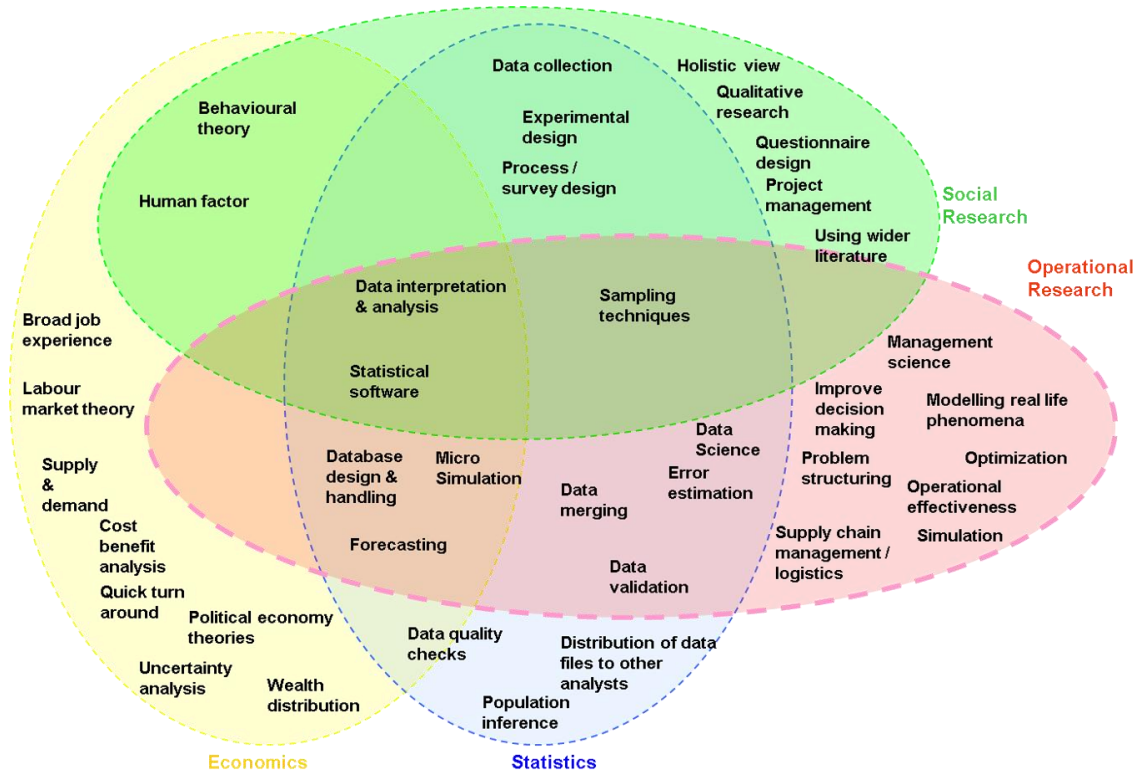
- Goal setting
- Active listening
- Managing input
- ... and behaviour

Follow-up

- Following up
- Writing up
- Monitoring impact



Stuff you already know!



Option	Cost	Impact	Deliverability
Option 1	£	!!!!	Amber
Option 2	££	!!!!!!!!	Green
Option 3	£££	!!!	Red

To structure a problem you must...

Define the problem	Context
	Analytical questions
	Relevant factors
Embrace the complexity	Address the whole problem
	Explore and eliminate ambiguity
	Add a useful structure
Be practical	No “magic wands”
	Consider data/evidence needed and how to obtain it
	What are the risks? Will it work?
Involve people	Problem “owner” always involved and usually others as well
	Give appropriate value to knowledge and opinion
	Be sparing with the call on their time
Generate an outcome	Ensure shared understanding of the issues
	Write it up in an understandable format
	Clearly specify the analysis and data required

Your task today

1. Read and discuss your assigned problem within your group
2. Produce a problem structure for it using suitable methods...

You should consider:

- What are the relevant analytical questions?
- What factors do we need to know about to answer them?
- Are there any relationships between these factors?
- What analysis would you carry out throw light on the problem?
- What data is required and how will you obtain it?
- What are the risks for your approach and how will you mitigate them?

3. Present your proposed approach back to the wider group
4. You will need to manage your time!

Rules

- 1) Think first
- 2) Predict before seeing results
- 3) No “right” model at the start

Scenario

A hospital is reviewing staffing levels in its Emergency Department (ED).

Recently:

- Waiting times have increased
- Staff feel under pressure
- Managers are concerned about costs

Patients arrive throughout the day needing treatment. Some cases are quick to resolve, while others require more time and attention.

The department is staffed by a team of doctors and nurses. Staff work together to assess and treat patients.

Hospital management want to understand:

- What changes might improve performance while controlling cost.

They are open to analysis but are not yet sure what kind of modelling or data is needed.

You have been asked to investigate.

Part A

Problem Structuring

Part A

Worksheet Part A

In groups discuss:

- What decisions might be made?
- What would we need to know?
- Sketch the system

Time: 20 minutes

Share back

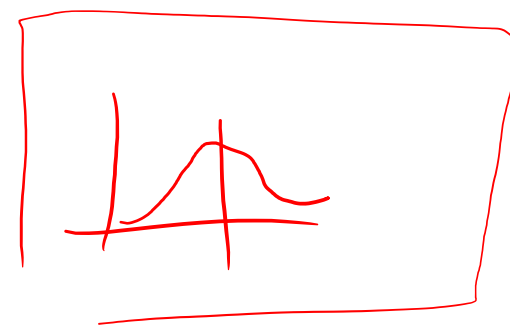
- Share what we have found from Part A

Time: 10 minutes

Decision

Reduce Staff pressure vs managing costs

Increasing staff $\rightarrow$ more throughput, less waiting time?



Analytical Qs

No doctors?

Money in? (Budget)

Average patient wait time?

Money out? (Costs)

Customers arriving per day?

Average service time

Highest demand dept.

Staffing roles (nurses)

Acceptable waiting time

Patient types

Peaktimes

$\rightarrow$ / Seasonal

times

Data

No doctors

Service time

$\rightarrow$ ~~cost~~ Patient arrival rate

Type of patient

Staffing Schedules

Part B

Prediction

Prediction Phase

Worksheet Part B

In groups discuss:

- What happens if staffing increases?
- What might still cause delays?
- What kinds of changes might improve the system?

↑ wait times ↓ optimal

Randomness of arrivals

Staff profile

Equipment not working
Amount of Equipment

Restocking System

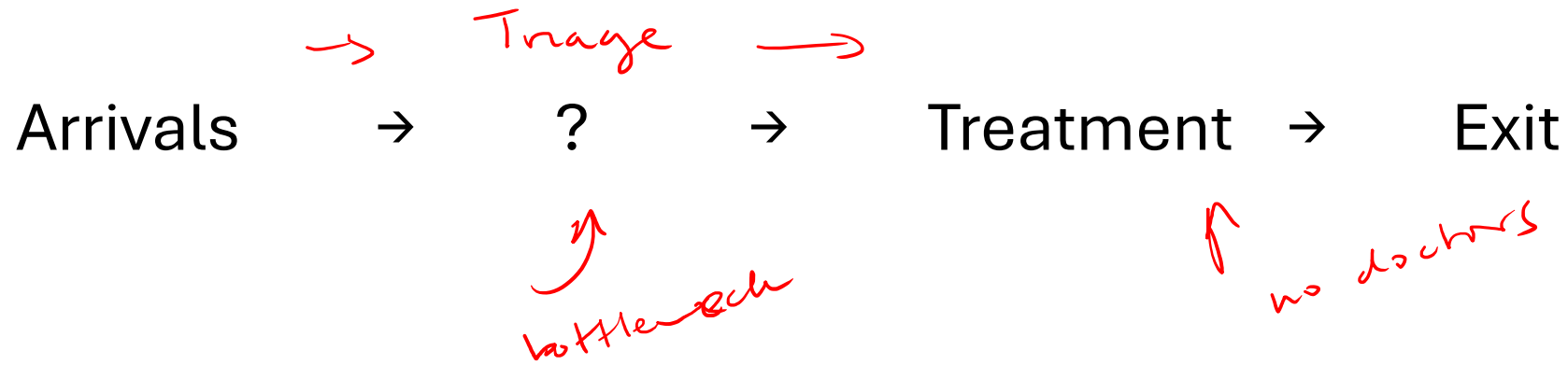
Staff training

Scheduling

Patient types

Time: 10 minutes

Conceptual Thinking

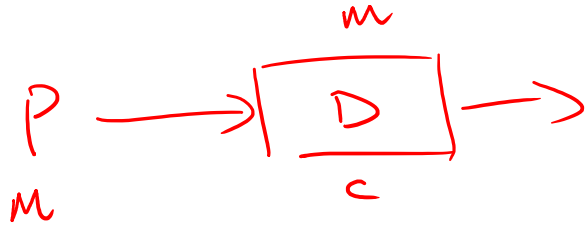


What might happen between arrival and treatment?

Part C

Building the Model

First Model

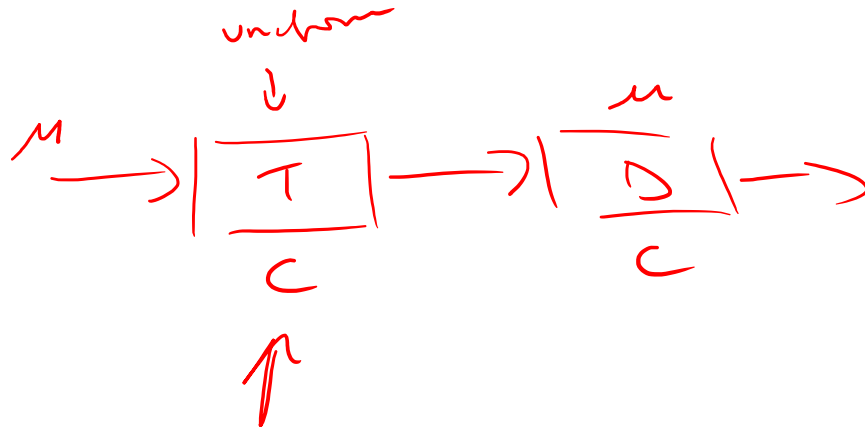


$m/m/c$

Add Triage

- Gather more information about a patient?

0 critical
1 urgent
2 standard



Variation

Some cases take longer than others

Add variability.

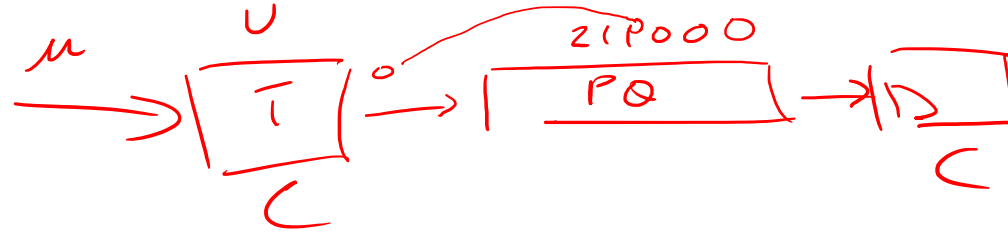
Patient Types!

- critical
- urgent
- standard

Fairness

Should everyone be treated strictly in arrival order?

Add in priority queue

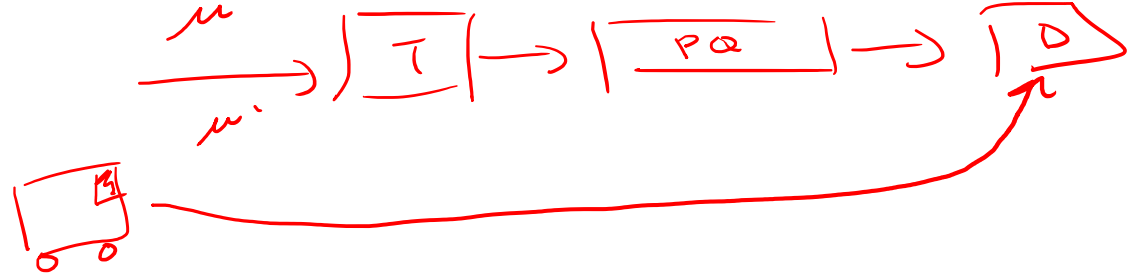


Extreme Case

What if something very serious arrives during treatment?

Add in pre-emption

Should anyone skip triage?



Major thing that we haven't modelled

Cost

We would require cost to do any serious analysis!

Part D

Results

What is the optimal number of doctors?

Define optimal:

- minimise critical breaches
- keep urgent breaches low
- avoid excessive cost
- maintain reasonable utilisation

You already have the data from the simulation. For each doctor level:

- critical breaches
- urgent breaches
- standard breaches
- utilisation

So, the decision rule becomes something like:

Choose the smallest number of doctors that keeps critical breaches ≈ 0 and urgent breaches acceptable.

Results from Simulation

You can access the full simulation that we have just built on the module.

Use this to fill out Part D and work out how many doctors are acceptable.

Also consider what costs would be needed for this simulation to be more realistic!